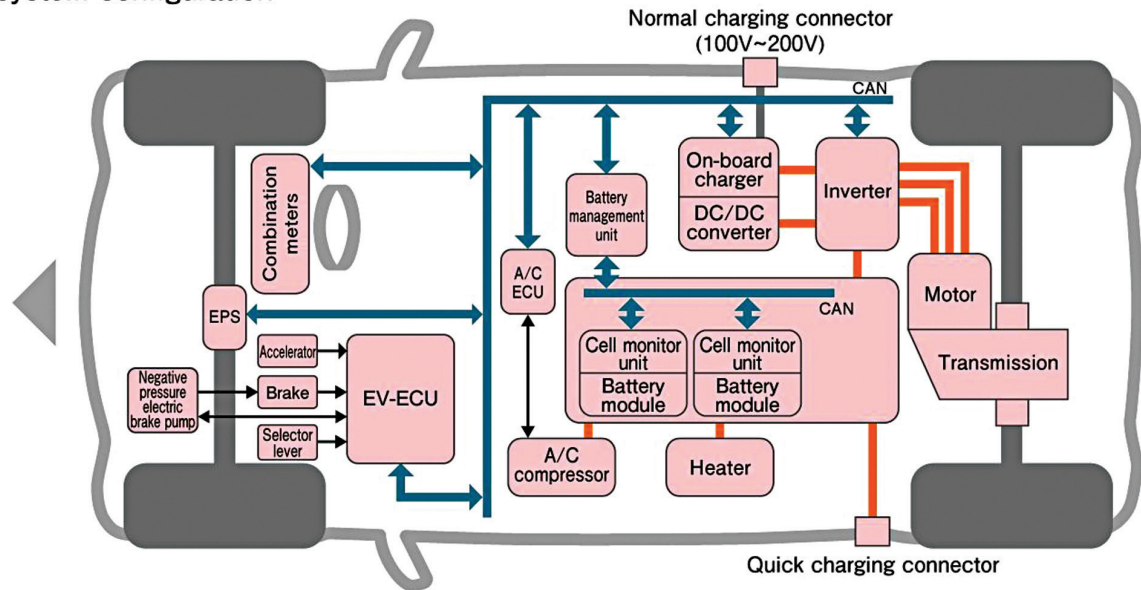


EV system configuration



Charging-to-driving process

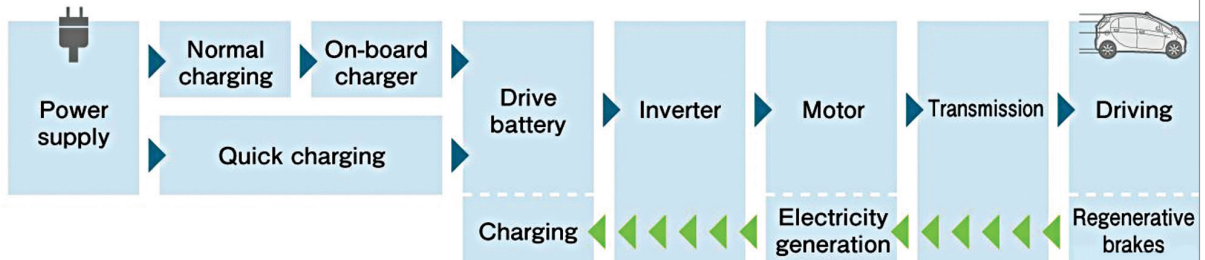


Fig. 3: Mitsubishi EV i-MiEV operational system configuration

one is the best, they both have their advantages and disadvantages. Different companies use different types of motors. For instance, Toyota uses AC induction motors, whereas Volkswagen prefers BLDC motors. Tesla Model S has an AC induction motor while Tesla Model 3 has a BLDC motor.

Controller. It is an electronic module that controls everything in a car—from the power deliverance of a battery to the power requirements of a motor.

Charger. There are two types of chargers. One is like a travel charger that can be plugged into a household electrical socket; this charger is generally slow. The other is the super-fast DC public chargers that are setup on the highways, fueling stations, or motels, etc.

Converter/Inverter. It can be DC-DC or DC-AC, depending upon the type of motor used (AC or DC).

Battery. Li-ion or Li-po (lithium-polymer) batteries are generally used in EVs due to their low maintenance and high power delivery characteristics.

EV system configuration

EV batteries are charged either through onboard chargers built within the cars from AC grid, or through fast chargers available at public charging stations. Onboard charging takes 7-8 hours to charge from 0% to 100%. But fast chargers installed in petrol pumps or highways can charge from 0% to 100% within 1-3 hours.

After the battery is charged, an inverter converts DC to AC or DC to DC depending on the voltage or current required by the motor.

There are regenerative brakes in most EVs, which generate electricity whenever brakes are applied and charge the batteries while driving.

The transmission consists of a gearbox and related components.

Most EVs have batteries in the car's base (due to safety reasons) instead of the boot or the front. If anything goes wrong with the battery, or it blasts, the car and the drivers will be safe. This is where battery management system (BMS) plays a critical role.

BMS with IoT & IoT

Now let us see the role of BMS in an EV, that is, how by integrating IoT and AI in BMS, we achieve another technological advancement.

Fig. 4 shows how typical integration of BMS with IoT and AI looks like. The multiple sensors are installed in the BMS of an EV to monitor voltage, current, temperature, and acoustics/vibration. These parameters are collected every split second. In intelligent vehicles and EVs, an edge intelligence system is